

Mini Data-Analysis: Deliverable 1

Kaizer Briggs

Total points available: 74

Part 0: Getting Set Up

Let's get ready to work on this assignment!

0.1: Install Packages

- Install the [diversedata](#) package by typing the following into your **R console**:

```
xinstall.packages("pak")  
library(pak)  
pak::pak("diverse-data-hub/diversedata")
```

0.2: Load Packages

Typically, R Packages are loaded in at the very beginning of the analysis. If you later want to use other packages, please come back and add them here:

```
library(tidyverse)  
library(diversedata)  
library(moderndiver)  
#--- Add any other packages below this line ---#
```

Task 1: Choose a Data Set and Research Question

You may use one of the datasets from class or one of the datasets from [diversedatahub](#).

- **boulder-housing:** This data set contains housing information for the Boulder, Colorado area. *[Add a second sentence here describing what the data covers — e.g., the variables included or what question it was collected to answer.]*
- **squirrel-census:** The [great NYC squirrel census](#), `squirrel-data.csv` – squirrel sightings recorded around Manhattan and Brooklyn parks.
- **rolling stone:** A [new visual essay](#) from The Pudding compares Rolling Stone’s “500 Greatest Albums of All Time” lists from 2003, 2012, and 2020. A methodology note says the project began with a spreadsheet by Chris Eckert and eventually led the authors to develop a dataset of their own. Theirs lists every album in the rankings — its name, genre, release year, 2003/2012/2020 rank, the artist’s name, birth year, gender, and more — plus each year’s voters. [h/t Jason Kottke]
- **coffee census:** In 2023, [British YouTuber](#) (and former [World Barista Champion](#)) James Hoffman virtually hosted the [Great American Coffee Taste Test](#), during which thousands of people simultaneously blind-tasted the same four coffees. Hoffman has published a [video summarizing the results](#), as well as a [spreadsheet of anonymized survey responses](#) from 4,000+ participants. It includes tasters’ demographics, general coffee drinking habits and preferences, assessments of the four coffees, and more. [h/t Dan Brady] (via [data-is-plural](#))
- **wildfire:** This data set contains information on wildfires in Canada, compiled from official government sources under the Open Government Licence – Alberta. The data was gathered to monitor, assess, and respond to wildfire risks across different regions. Wildfires have far-reaching environmental, social, and economic consequences. From an equity and inclusion perspective, analyzing wildfire data can reveal geographic and resource-based disparities in detection and containment efforts, and highlight how certain populations face greater risks due to climate change and limited infrastructure. There are 26551 rows and 35 columns.
- **genderassessment:** Collected in 2023, the data allows for comparative evaluation across countries, sectors, and ownership types (e.g., Public, Private, Government). Each record represents a company and its corresponding evaluation across 28 detailed gender related indicators, offering a comprehensive snapshot of corporate gender equity worldwide. There are 2000 rows and 29 variables
- **hcmst:** This data set is adapted from the original data set [How Couples Meet and Stay Together 2017, 2022](#). This study, led by researchers from Stanford University, surveyed 1,722 U.S. adults in 2022 to explore how relationships form and change with time and focused on dating habits and the impact of the COVID-19 pandemic on relationships. This adapted data set focuses on variables that may affect the quality of the relationship,

considering demographic characteristics of the subjects, couple dynamics, as well as COVID-19-related variables. The COVID-19 pandemic had a [significant impact](#) on romantic relationships in the United States. This data set enables exploration of how external factors, like the health of the subjects and changes in income, as well as personal behaviors, like conflict and intimate dynamics, relate to an individual's perception of the quality of the relationship. There are 1328 rows and 21 columns.

- **womensmarchmadness:** This adapted data set contains historical records of every NCAA Division I Women's Basketball Tournament appearance since the tournament began in 1982 up until 2018, capturing tournament results across more than four decades of collegiate women's basketball. All data is sourced from the NCAA and contains the data behind the story [The Rise and Fall Of Women's NCAA Tournament Dynasties](#). The rise in popularity of the NCAA Women's March Madness, fueled by athletes like Caitlin Clark and Paige Bueckers, reflects a broader cultural shift in the recognition of women's sports. Beyond entertainment and athletic achievement, women's participation in sport has social and professional benefits. There are 2092 rows and 20 columns.

Note: We encourage you to use one of the options above, but if you have a data set that you'd really like to use, please check with a member of the teaching team to see whether the data set is of appropriate complexity. If approved, please add a brief description of the data here.

1.1: Choose 2 data sets (2 points)

Out of the 5 data sets listed above, choose **2** that appeal to you based on their description. Write your choices below:

1: Write 1st choice here Boulder 2: Write 2nd choice here Squirrel

1.2: Explore the Data (12 points)

One way to narrowing down your selection is to *explore* the data sets. Use your knowledge of `dplyr` to summarize three variables in each of the data sets (for example, listing what levels of a categorical variable exist, or calculating the mean of a continuous variable of interest). Write a sentence that describes your findings for each variable explored. You may use multiple R code chunks if preferred.

Data Set 1

```
### Explore 3 variables of data set 1 ###  
  
library(tidyverse)  
boulder <- read_csv("dat/boulder-2020-residential_sales.csv")
```

Rows: 3052 Columns: 37

-- Column specification -----

Delimiter: ","

chr (24): Account #, PARCELNB, PROPERTY_ADDRESS, LOCCITY, SUBNAME, MULTIPLE_...

dbl (13): Market

Area 1, BLDG1_YEAR_BUILT, BEDROOMS, FULL_BATHS, THREE_QTR_B...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
# Explore sale price
boulder %>%
  summarize(
    mean_sale_price = mean(
      as.numeric(str_remove_all(SALE_PRICE, "$,")),
      na.rm = TRUE
    )
  )
```

```
# A tibble: 1 x 1
  mean_sale_price
      <dbl>
1      652538.
```

```
# Explore number of bedrooms
boulder %>%
  summarize(
    mean_bedrooms = mean(BEDROOMS, na.rm = TRUE),
    min_bedrooms = min(BEDROOMS, na.rm = TRUE),
    max_bedrooms = max(BEDROOMS, na.rm = TRUE)
  )
```

```
# A tibble: 1 x 3
  mean_bedrooms min_bedrooms max_bedrooms
      <dbl>         <dbl>         <dbl>
1      3.22          0          16
```

```
# Explore account type
boulder %>%
  count(ACCOUNT_TYPE)
```

```
# A tibble: 2 x 2
  ACCOUNT_TYPE      n
  <chr>          <int>
1 RESIDENTIAL    2556
2 RESIDENTIAL CONDO 496
```

Write your findings here. The Boulder housing dataset contains 3,052 residential property sales and 37 variables. The average sale price is approximately \$652,538. The number of bedrooms ranges from 0 to 16, with an average of 3.22 bedrooms per property. The dataset contains 2,556 RESIDENTIAL properties and 496 RESIDENTIAL CONDO properties.

Data Set 2

```
### Explore 3 variables of data set 2 ###
squirrel <- read_csv("dat/squirrel-data.csv")
```

Rows: 433 Columns: 16

-- Column specification -----

Delimiter: ","

chr (14): Area Name, Area ID, Park Name, Park ID, Squirrel ID, Primary Fur C...

dbl (2): Squirrel Latitude (DD.DDDDDD), Squirrel Longitude (-DD.DDDDDD)

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
squirrel
```

```
# A tibble: 433 x 16
```

	`Area Name`	`Area ID`	`Park Name`	`Park ID`	`Squirrel ID`	`Primary Fur Color`
	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
1	UPPER MANH~	A	Fort Tryon~	01	A-01-01	Gray
2	UPPER MANH~	A	Fort Tryon~	01	A-01-02	Gray
3	UPPER MANH~	A	Fort Tryon~	01	A-01-03	Gray
4	UPPER MANH~	A	Fort Tryon~	01	A-01-04	Gray
5	UPPER MANH~	A	Fort Tryon~	01	A-01-05	Gray
6	UPPER MANH~	A	Fort Tryon~	01	A-01-06	Gray
7	UPPER MANH~	A	Fort Tryon~	01	A-01-07	Gray
8	UPPER MANH~	A	Fort Tryon~	01	A-01-08	Black
9	UPPER MANH~	A	Fort Tryon~	01	A-01-09	Gray
10	UPPER MANH~	A	Fort Tryon~	01	A-01-10	Gray

```
# i 423 more rows
# i 10 more variables: `Highlights in Fur Color` <chr>, `Color Notes` <chr>,
#   Location <chr>, `Above Ground (Height in Feet)` <chr>,
#   `Specific Location` <chr>, Activities <chr>,
#   `Interactions with Humans` <chr>, `Other Notes or Observations` <chr>,
#   `Squirrel Latitude (DD.DDDDDD)` <dbl>,
#   `Squirrel Longitude (-DD.DDDDDD)` <dbl>
```

```
# Explore primary fur color
squirrel %>%
  count(`Primary Fur Color`)
```

```
# A tibble: 4 x 2
  `Primary Fur Color`     n
  <chr>                 <int>
1 Black                  16
2 Cinnamon               26
3 Gray                   390
4 <NA>                    1
```

```
# Explore location
squirrel %>%
  count(Location)
```

```
# A tibble: 7 x 2
  Location                                     n
  <chr>                                     <int>
1 Above Ground                             84
2 Above Ground, Specific Location          54
3 Ground Plane                           220
4 Ground Plane, Above Ground              10
5 Ground Plane, Specific Location          7
6 Specific Location                       24
7 <NA>                                    34
```

```
# Explore squirrel latitude
squirrel %>%
  summarize(
    mean_latitude = mean(
      `Squirrel Latitude (DD.DDDDDD)`,
      na.rm = TRUE
    )
  )
```

```

    ),
    min_latitude = min(
      `Squirrel Latitude (DD.DDDDDD)` ,
      na.rm = TRUE
    ),
    max_latitude = max(
      `Squirrel Latitude (DD.DDDDDD)` ,
      na.rm = TRUE
    )
  )
)

```

```

# A tibble: 1 x 3
  mean_latitude min_latitude max_latitude
      <dbl>         <dbl>         <dbl>
1      40.8         40.7         40.9

```

Write your findings here.

The squirrel census dataset contains 433 squirrel observations and 16 variables. Gray was the most common primary fur color, with 390 squirrels, followed by Cinnamon with 26 and Black with 16. One observation did not have a recorded primary fur color. Ground Plane was the most common location category, with 220 observations, followed by Above Ground with 84 observations. There were also 34 observations with a missing location. The average squirrel latitude was approximately 40.8, with recorded latitudes ranging from approximately 40.7 to 40.9.

1.3: Choose 1 Data Set (2 points)

It's time to choose only one data set. State the data set that you've chosen, and why you've chosen it.

I chose the Boulder housing dataset because it has a large number of observations and several quantitative and categorical variables that can be used to investigate relationships between characteristics of residential properties and their sale prices. The dataset also has variables such as sale price, number of bedrooms, and account type that can be used to develop specific research questions for the rest of the analysis.

1.4: Research Question (4 points)

Let's choose a primary and a secondary research question to explore.

Write your research questions **as questions**, and be specific. You can change it later if needed.

For example, if I had chosen a `titanic` data set for my project, I might ask,
“(Primary) Is there a relationship between survival and the class of the passengers?
(Secondary) Does this relationship differ by gender?”

Primary: Is there a relationship between the number of bedrooms and the sale price of residential properties in the Boulder housing dataset?

Secondary: Does the relationship between the number of bedrooms and sale price differ between residential properties and residential condominiums?

1.5: Commit (2 points)

Commit your work and push it to GitHub. Include an informative commit message, and include “(1.5)” in the message.

Task 2: Further Exploring Your Chosen Data Set

2.1: Missing Data (6 points)

Missing data is inevitable, and can complicate analyses. Let’s see what variables (if any) have missing data in your chosen data set.

Your task is to create a table that calculates the proportion of missing values per variable. Be sure to output the table.

```
### Explore missingness here ###
boulder %>%
  summarize(
    across(
      everything(),
      ~ mean(is.na(.))
    )
  ) %>%
  pivot_longer(
    cols = everything(),
    names_to = "variable",
    values_to = "proportion_missing"
  ) %>%
  arrange(desc(proportion_missing))
```

```
# A tibble: 37 x 2
  variable                proportion_missing
  <chr>                    <dbl>
1 "MAILING_ADDR2"         0.999
2 "CARE_OF"               0.996
3 "Grantor 1"             0.000983
4 "Market\rArea 1"        0
5 "Account #"             0
6 "PARCELNB"              0
7 "PROPERTY_ADDRESS"      0
8 "LOCCITY"               0
9 "SUBNAME"               0
10 "MULTIPLE_BLDGS"        0
# i 27 more rows
```

2.2: Missing Data (Again) (6 points)

Based on your research question, will this missingness pose an issue? For the purposes of this class (and this class only!), we will consider missingness a problem **if there is more than 20% of a single variable (that is of interest) is missing.**

For example, let's assume I wanted to explore the following research questions: "Is there a relationship between survival and the class of the passengers? Does this relationship vary by gender?". If the variable indicating whether or not a person survived was missing for 20% or more of the passengers, then this would be a problem. However, if a variable indicating the colour of shirt a passenger was wearing was missing, this probably wouldn't be an issue as that variable is quite irrelevant to my analysis!

Based on this definition, is missingness an issue for your analysis? If so, describe how you will address this (pivoting your research question, for example). If you will continue with a new research question, write it here! **Do not go back to Task 1 and redo the analysis.**).

If missingness is not an issue, describe why.

Missingness is not an issue for my analysis because the variables that are directly related to my research questions do not have more than 20% missing values. In particular, the variables for sale price, number of bedrooms, and account type have sufficient data for the analysis. Although some other variables in the Boulder dataset have missing values, they are not relevant to my research questions. Therefore, I will continue with my original research questions.

2.3: Tidy your Data (10 points)

Produce a tidy data set that could be used to answer your research questions. **Please ensure you have at least one quantitative (numeric) and one categorical variable in your data set. It's okay you need to include a less relevant variable in your tidied data to ensure this.**

To tidy your data, you should:

- Create new variables (if needed)
- Transform the data into a tidy form (if needed)
- Remove irrelevant columns (if needed)
- Comment your code throughout

Show the first 6 rows of the tidied data.

```
boulder_tidy <- boulder %>%  
  
  mutate(  
    sale_price = as.numeric(str_remove_all(SALE_PRICE, "$,") )  
  ) %>%  
  
  select(  
    sale_price,  
    bedrooms = BEDROOMS,  
    account_type = ACCOUNT_TYPE  
  )  
  
# Show the first 6 rows of the tidied dataset  
boulder_tidy %>%  
  slice_head(n = 6)
```

```
# A tibble: 6 x 3  
  sale_price bedrooms account_type  
    <dbl>    <dbl> <chr>  
1   1750000         4 RESIDENTIAL  
2   1842000         5 RESIDENTIAL  
3   1200000         3 RESIDENTIAL  
4    610000         4 RESIDENTIAL  
5    696900         6 RESIDENTIAL  
6    654400         5 RESIDENTIAL
```

2.4: Create a Table (10 points)

Use any functions from the `tidyverse` to create one table that outputs the mean, minimum, and maximum of all numeric columns in your data, dropping the missing values if they exist.

Show the outputted table.

```
boulder_tidy %>%
  summarize(
    across(
      where(is.numeric),
      list(
        mean = ~ mean(.x, na.rm = TRUE),
        min = ~ min(.x, na.rm = TRUE),
        max = ~ max(.x, na.rm = TRUE)
      )
    )
  )
```

```
# A tibble: 1 x 6
  sale_price_mean sale_price_min sale_price_max bedrooms_mean bedrooms_min
      <dbl>         <dbl>         <dbl>         <dbl>         <dbl>
1    652538.         10000         6600000         3.22          0
# i 1 more variable: bedrooms_max <dbl>
```

2.5: Commit (2 points)

Commit your work and push it to GitHub. , and include “(2.7)” in the message.

Task 3: Tidy Your Submission Overall

Check over your document and GitHub repository for the following:

3.1: Coherence (2 points)

The document should read sensibly from top to bottom, with no major continuity errors. An example of a major continuity error is having a data set listed for Task 3 that is not part of one of the data sets listed in Task 1.

3.2: Error-free code (2 points)

For full marks, all code in the document should run without error and be completely reproducible.

3.3 README (6 points)

There should be a file named `README.md` at the top level of your repository. Its contents should automatically appear when you visit the repository on GitHub.

Minimum contents of the README file:

- In a sentence or two, explains what this repository is, so that future-you or someone else stumbling on your repository can be oriented to the repository.
- List the files/folders contained in the repository
- In a sentence or two, briefly explains how to engage with the repository. You can assume the person reading knows the material from STAT 545A. Basically, if a visitor to your repository wants to explore your project, what should they know? How can they reproduce your report?

3.4 Generative AI Disclosure (3 points)

In this course, Generative AI can be used in the following ways:

- to clarify concepts discussed in class
- as an “advanced search engine” (i.e., searching error codes)
- debugging code that students wrote and attempted to debug on their own

Generative AI **CANNOT** be used to generate text or code (including comments) from scratch.

Any use of Generative AI must be disclosed.

To disclose your use, please copy and paste the following template into the README of your GitHub Repository and fill out the relevant details [in square brackets]. BE SPECIFIC. Saying you used it to debug your code is not enough. Explicitly describe where you got stuck

Here is an example of a specific, explicit debug:

“I had the error `attempt to apply non-function` after running my code. I used Claude to help me identify that this error was due to me attempting to multiply two numbers together without the use of a `*`, i.e. `(2)(3)` instead of `2*3`.”

```
## Generative AI Statement
```

```
Generative AI (through [LIST MODELS USED, i.e. ChatGPT, CoPilot]) was used to help me complete
```

1. Claude Code was used to help troubleshoot an error when importing the Boulder dataset. It
 2. Claude Code was used to help troubleshoot GitHub and Positron synchronization issues, including
- ...

```
I affirm that Generative AI was not used to generate text, code, or comments for my assessments.
```

If you did not use Generative AI, please include the following in your README:

```
## Generative AI Statement
```

```
Generative AI was not used in any way throughout this assignment.
```

Assessments suspected of having AI-generated text and/or code, or assignments where the Generative AI use was not disclosed, will be flagged and temporarily assigned a grade of zero. Students will be required to meet with the instructor to receive a grade.

3.5 Output (4 points)

All output on GitHub is readable, recent and relevant:

- All `.qmd` files have been rendered to their output `.md` files.
- All rendered `.md` files are viewable without errors on Github. Examples of errors: Missing plots, “Sorry about that, but we can’t show files that are this big right now” messages, error messages from broken R code
- All of these output files are up-to-date – that is, they haven’t fallen behind after the source (`.qmd`) files have been updated.
- There should be no relic output files. For example, if you were rendering a `.qmd` to `.html`, but then changed the output to be only a markdown file, then the `.html` file is a relic and should be deleted.

Step 4: Submission

**** Submit repo link ****

To submit this milestone, submit the github link to the repo.

This assignment was authored by the team of instructors at University of British Columbia's STA 545 class.